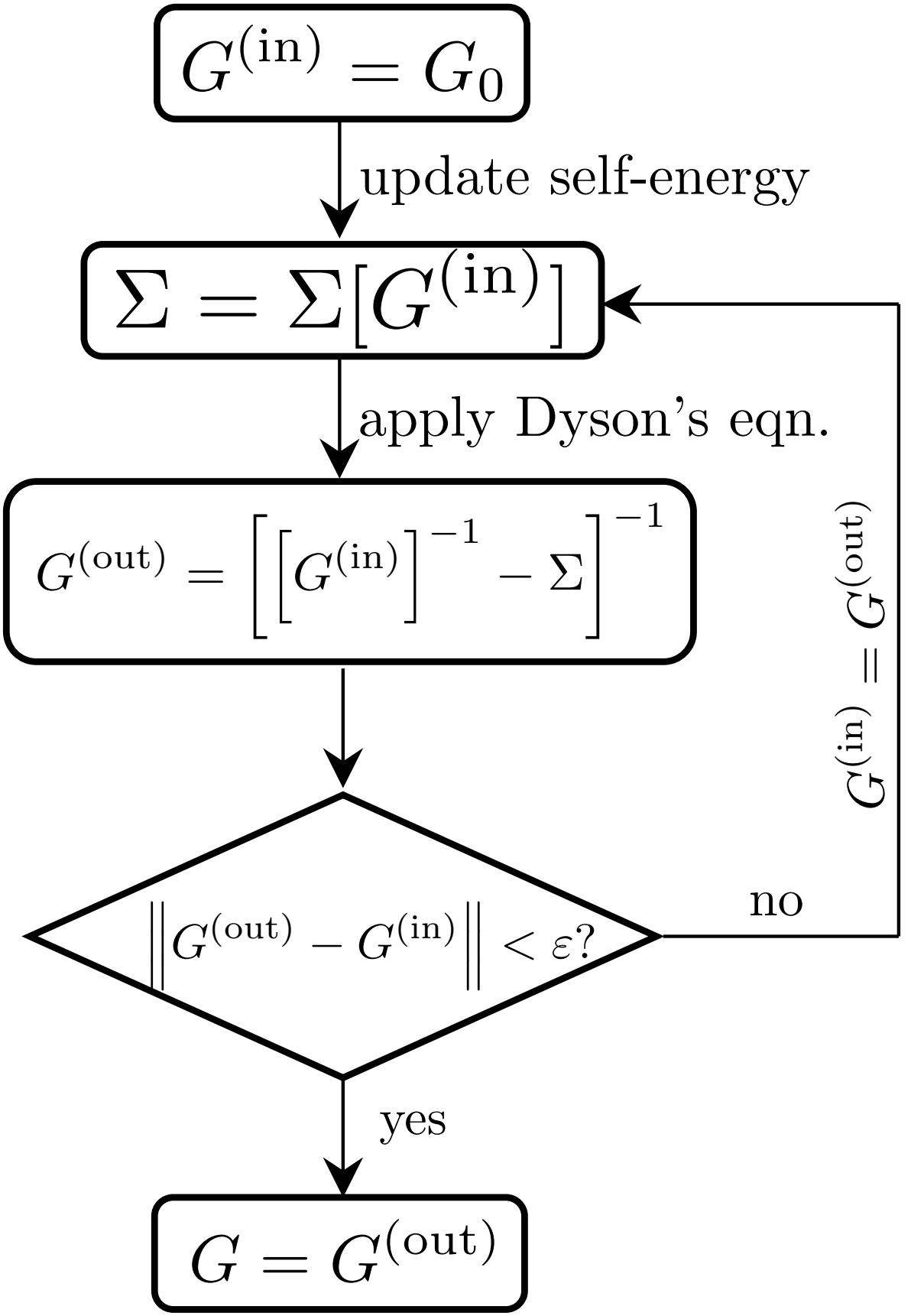

$G^{(\mathrm{in})} = G_0$
update self-energy
$\Sigma = \Sigma[G^{(\mathrm{in})}]$
apply Dyson's eqn.
$G^{(\mathrm{out})} = \left[\left[G^{(\mathrm{in})}\right]^{-1} - \Sigma\right]^{-1}$
$\left\|G^{(\mathrm{out})} - G^{(\mathrm{in})}\right\| < \varepsilon?$
no
$G^{(\mathrm{in})} = G^{(\mathrm{out})}$
yes
$G = G^{(\mathrm{out})}$